Domestic hot water (DHW) module — functional description

Supplies hot water to the taps from three buffer tanks, maximising the use of heat recovered from the drives, DC and high-temperature circuits.

What it does

Each tank is in exactly one state: **in use**, **filling**, **boosting**, **needs fill**, **needs boost**, **standby** or **disabled**. The module reassesses all three once a second, keeping one tank in use, at most one filling and at most one boosting.

Fresh water entering a filling tank is preheated on the way in: flow-control valves on the drives recovery and DC circuits modulate to hold their return lines at `filling_temperature_setpoint`. The drives branch closes when the drives return no recoverable heat.

Boosting circulates a full tank that has dropped below `minimum_tank_temperature` through a loop fed by one of two heat sources, at fixed flow, until it reaches `maximum_tank_temperature`. The pump starts only once the source valve and a tank supply-and-return pair read open — they take about 90 seconds to travel — so it never runs against a closed circuit.

When no tank is full and at or above `minimum_tank_temperature`, supply continues at reduced temperature: the tank in use stays in service while below temperature, unless a warmer tank is available, in which case the warmest tank still holding water is selected. A warning is raised in both cases. When neither heat source is available, boosting remains idle and the tanks cool until a source returns.

Modes

The panel reports two modes for this module, independently:

- **Filling** — *idle* or *filling*.
- **Boosting** — *idle*, *high-temperature boosting* or *heatpump boosting*.

Boosting

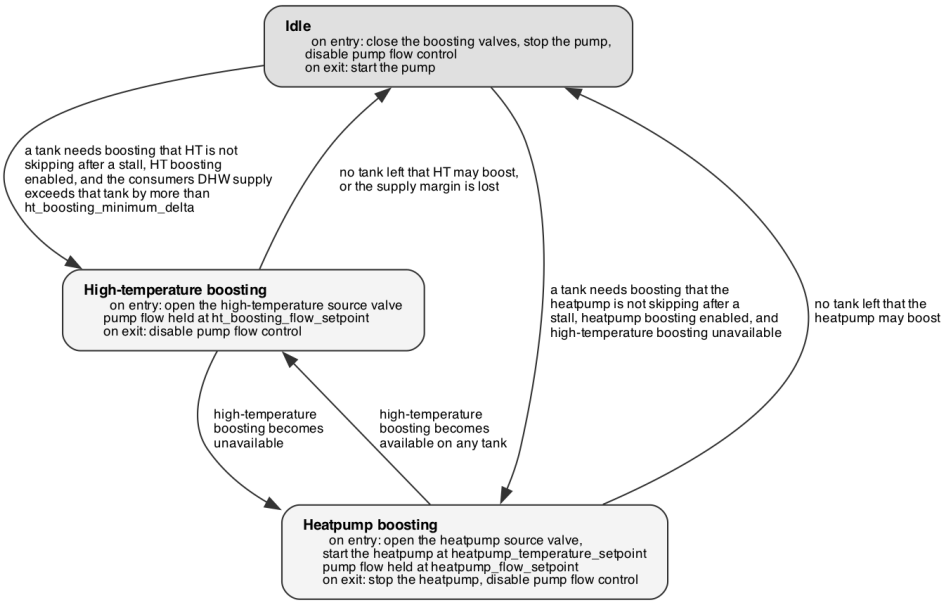


Figure 1: DHW boosting mode

The high-temperature loop is the preferred source and the heatpump the fallback: heatpump boosting is available only while high-temperature boosting is not, and the module moves directly between the two without passing through idle. High-temperature boosting takes over from the heatpump as soon as any tank qualifies for it, also when that is not the tank the heatpump is heating.

While boosting, the module compares heat transfer into the tank against `boosting_minimum_heat`. When it stays below that threshold for longer than `boosting_stall_window`, after an initial `boosting_startup_grace`, the source in use skips that tank for `boosting_stall_lockout`; the other source may still boost it. In the same second the module boosts another tank from the same source, switches to the other source, or goes idle. The skipped tank still shows *needs boost*. The startup grace restarts whenever the source or the tank changes.

Tuning

Sources

- **`ht_boosting_enabled`, `heatpump_boosting_enabled`** — on, off. Which sources may be selected. Enable the heatpump to continue boosting when the high-temperature loop is cold; disable a source to exclude it entirely.
- **`ht_boosting_minimum_delta`** — 3 K. How far the consumers' DHW supply must exceed the tank temperature before high-temperature boosting starts or continues. Raise it if boosting starts on a marginal supply and stops on the heat-transfer threshold; lower it to keep using the loop at smaller margins.
- **`ht_boosting_flow_setpoint`** — 10 l/min. Boosting loop flow from the high-temperature source, set by the rating of the DHW exchanger.
- **`heatpump_flow_setpoint`** — 25 l/min. Boosting loop flow from the heatpump, set by the heatpump's rated flow.
- **`heatpump_temperature_setpoint`** — 65 °C. Supply temperature commanded from the heatpump while boosting.

Tank temperature and level

- **`minimum_tank_temperature`** — 45 °C. Serves two purposes: a full tank below it is boosted, and a tank below it is too cold to be selected for use. Raise it for hotter water at the taps and more frequent boosting; lower it to boost less often.
- **`maximum_tank_temperature`** — 60 °C. Where boosting stops. Raise it to store more heat per tank and boost less often; lower it to reduce standing losses and scalding risk. Must stay above `minimum_tank_temperature`.
- **`maximum_tank_level`** — 230 L. Where filling stops; the accepted maximum is 275 L and the high-level warning is at 265 L.
- **`full_level_lower_band`** — 200 L. The level above which a tank counts as full, and so the level at which filling restarts. Widen the gap to `maximum_tank_level` to refill less often; narrow it to keep tanks nearer full.
- **`minimum_tank_level`** — 30 L. The level at which a tank in use counts as empty and is swapped out. Raise it to swap earlier; lower it to draw more from each tank.
- **`filling_temperature_setpoint`** — 40 °C. The return-line temperature the drives and DC preheat valves hold while a tank fills. Raise it to recover more heat during filling, leaving less for boosting; lower it if the recovery circuits are pulled down too far.

Availability and guards

- **`tank1_enabled`, `tank2_enabled`, `tank3_enabled`** — all on. Take a tank out of service. A disabled tank is released immediately, without waiting for its valves to travel.
- **Stall guard** (`boosting_minimum_heat` 1000 W, `boosting_stall_window` 120 s, `boosting_startup_grace` 180 s, `boosting_stall_lockout` 900 s) — the heat transfer threshold below which a tank counts as stalled, how long it must stay below, the delay after boosting starts on a source or tank before checking starts, and how long the source then skips that tank. Lengthen the grace and window if boosting stalls during normal warm-up. Shorten the lockout to retry a stalled tank sooner; lengthen it if a tank keeps stalling on the same source.
- **Minimum valve and pump openings** (`minimum_pump_dutypoint` 0.3, `drives_flowcontrol_minimum_setpoint` 0.1, `dc_flowcontrol_minimum_setpoint` 0.1) — floors that guarantee flow through the system, for example so the temperature sensors read the circulating medium.
- **Gains for the PID controllers** (`pump_flow_tuning`, `dc_flow_tuning`, `drives_flow_tuning`) — proportional, integral and derivative gains, tuned during commissioning.

The module rejects a parameter set outright, with nothing applied, if the maximum tank temperature is below the minimum, if the maximum level is below the minimum, or if `full_level_lower_band` falls outside the level band. If a change does not take effect on the panel, check these first.

Alarms

Alarm	Severity	Raised when
Tank n high temperature warning / alarm	Warning / Alarm	A tank reads more than 2 K (warning) or 5 K (alarm) above `maximum_tank_temperature`. Boosting should have stopped; check the source valves and the tank sensor.

Alarm	Severity	Raised when
Tank n high level warning / alarm	Warning / Alarm	A tank reads above 265 L (warning) or 270 L (alarm). Filling should have stopped; check the inlet valve.
Tank in use below minimum temperature	Warning	The tank in use is below `minimum_tank_temperature` and no warmer tank was available to take over. Supply continues at reduced temperature.

Derived from `src/thrs/control/modules/dhw.py`. Re-issue this manual when the control logic changes.